Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years shipping **LLM-powered** products across the stack in **Python** and **TypeScript**, from multimodal document pipelines to **citation-grounded retrieval** and **secure multi-tenant** data systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Restructured a fault-tolerant async **LLM orchestration** system on FastAPI background workers with explicit job-state transitions and failure isolation, processing 1K+ AI jobs and tripling throughput under concurrent load.
- Architected a multi-tenant AI education platform on Azure SQL and Blob Storage with **row-level security** isolating tenant data for 500+ users, tuning indexed queries to cut dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built a **document intelligence** pipeline ingesting PDF, DOCX, and ZIP submissions through OCR and multimodal LLMs (GPT-4o, Claude, Qwen-VL), grading 10K+ daily requests for 150+ instructors and cutting feedback turnaround 60%.
- Placed OpenAI, Anthropic, and Qwen models behind a switchable provider layer with Claude Vision fallback for OCR timeouts and per-call token cost logging, enabling **cost-aware model selection** across grading workflows.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built a GPT-based **Decision Transformer** for offline RL over 100K+ driving trajectory logs, improving long-horizon planning 18% in simulation and comparing it against Wayformer-style and MotionLM-style encoders.
- Designed a spatial-temporal encoder fusing HD-map geometry with ego-agent trajectory history through **multi-head attention**, improving closed-loop planning stability 15% in experiments tracked with Hydra and Weights & Biases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built a resource-management platform in **TypeScript**, React, and Node.js across eight microservices with WebSocket real-time updates on PostgreSQL, synchronizing 2M+ records across locations for allocation, billing, and lead workflows.
- Automated GitHub Actions CI/CD with unit, integration, and end-to-end tests, Docker packaging, and zero-downtime releases, then used product analytics to cut resource wastage 42% and raise lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **FastAPI** booking, pricing, coupon, and payment services on PostgreSQL with WebSocket status push, holding sub-200ms latency across 50K+ records and speeding booking confirmations 40% for mobile and web.
- Automated idempotent payment webhooks for refunds and disputes, cutting manual refund handling 70%, and **presented the end-to-end architecture** to client stakeholders before build, reducing design errors 25%.

Skills

Languages & Web: Python, TypeScript, JavaScript, Java, SQL, C++, React.js, Next.js, Node.js
Backend: FastAPI, Flask, REST APIs, WebSockets, Microservices, Distributed Systems, System Design
AI & Retrieval: LLM Systems, RAG, BM25, Hybrid Search, FAISS, Pinecone, Embeddings, Prompt Engineering, OCR
Data & Cloud: PostgreSQL, Azure SQL, MongoDB, Docker, Kubernetes, AWS, Azure, GCP, GitHub Actions, CI/CD
Security & ML: Row-Level Security, Access Control, Multi-Tenancy, Retrieval Evaluation, Transformers, PyTorch

Projects

ClauseTrace - Citation-Grounded Contract Retrieval | *Python, BM25, Rank Fusion, FastAPI, Claude API*

- Built a **citation-grounded** contract retrieval engine fusing BM25 and n-gram rankings with learned clause profiles, lifting hit@1 38% on 1,244 held-out CUAD queries with zero citation mismatches or cross-matter leaks.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Convex, Clerk, GCP*

- Built an AI document workspace with LLaMA-2 and FAISS **semantic search**, summarization, Clerk-secured uploads, and Convex real-time collaboration, reaching 95% query accuracy across 50K+ documents for 200+ users.

Chat2SQL Code | *TypeScript, Java, SQL, PostgreSQL, Docker*

- Extended an **AI-assisted** SQL workspace in TypeScript and Java that turns natural-language intent into schema-aware queries across 10+ databases, validating each query before execution at 82% generation accuracy.